

FPGA Hackathon 2026 – Quick Reference Guide

Getting Started

Vivado version: 2025.2

To launch your project from Vivado's Tcl Console:

```
cd <path_to_project>/vivado/  
source hackathon.tcl
```

Project Structure and Configuration

- **siemens/, synopsys/** - contains the files needed to complete the task 4 and 6
- **src/** – contains control modules; modify only `task_enabler_pkg.sv`.
- **src/task_wrappers/** – includes wrappers for input/output handling; do not modify.
- **src/tasks/task_x/** – each folder holds a task implementation file; you may add supporting files.
- **tb/** – testbench and .mem files for local simulation (unless specified otherwise).
- **vivado/** – Tcl script and project files generated by Vivado.

Enable implemented tasks with 1'b1 in `task_enabler_pkg.svh`. All implemented tasks must be enabled at the same time, as the judge tests them in a single bitstream run and the score depends on how many work correctly. You can use any HDL language you prefer.

Uploading Your Solution

1. Remove all *.dcp files
 2. Generate Tcl script:
File > Project > Write Tcl...
Uncheck: Write all properties, Copy sources, Recreate Block Designs
 3. Ensure section 2 of `hackathon.tcl` is <none>
 4. Commit `hackathon.tcl` + updated source files to the main branch
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Task Evaluation

Each time you trigger a job on Jenkins, your solution will first be tested in **Development mode** for debugging and verification purposes, and immediately afterwards in **Evaluation mode** for final scoring:

Development Mode	Evaluation Mode
Single test vector per task	Multiple fixed test vectors per task
Randomized	Same for all teams
Logs available	Logs not available
Use for debugging (not scored)	Use for scoring (only the latest run)

Task Scoring Table

Task number	Task name	Task file	Evaluation mode scoring	Maximum amount of bonus points	Theoretical total points*
1	Shifter	task_1.sv	8	5	13
2	Self-Calibration [HITACHI]	task_2.vhd	30	10	40
3	Morse	task_3.sv	10	5	15
4	Frame Extender [SYNOPSYS]	synopsys/	10	N/A	10
5	Enigma Encryption	task_5.sv	15	5	20
6	Packet Parser [SIEMENS]	siemens/	6	4	10
7	Digital controller for a bionic arm [ONT]	task_7.sv	16	5	21
8	Software task	task_8.sv	20	N/A	20
9	Cracovian	task_9.sv	15	5	20
10	Memory Mapped	task_10.sv	15	5	20
11	Artifacts	task_11.sv	15	5	20
12	Abacus Romanus	task_12.sv	12	5	17
13	NCO	task_13.sv	12	5	17
14	MCU	task_14.sv	15	5	20
15	LDPC Encoder	task_15.sv	15	5	20
16	AXI Stream Compressor	task_16.sv	12	5	17

* This is a theoretical value due to latency and resource utilization optimization trade-offs.

You can only edit the main task files listed in the table - do not rename them or change their interfaces. You may add supporting modules or files under the relevant task_x folder.

Bonus Points

Some tasks offer additional points for **resource utilization**. Bonus points are awarded only if the task passes all test vectors in Evaluation Mode. For some tasks, the number of bonus points may also depend on the achieved latency value, and bonus points are granted only if the latency is below the maximum allowed threshold. More information in the FPGA_Hackathon_2026_Project_Guide.pdf and in the task-specific instructions.

Available Simulators

The project includes two testbenches: tb_top and tb_lite (a faster version). You can run local simulations using **Vivado Simulator**, **QuestaSim/Visualizer**, or **VCS/Verdi** - choose whichever you prefer for testing your solution.

Tips

If you use VM, we recommend using only **1 job for synthesis** (more jobs may slow-down the machine or even freeze it).